

The Pentose Phosphate Pathway as a Two-Module, Mode-Switching Central-Carbon Device

*A commissioned review-style synthesis for a molecular-biology audience. *Pseudomonas putida* KT2440 proteins are used as concrete bacterial exemplars, but the intended boundary of the system is the conserved chemistry of glucose-6-phosphate oxidation and pentose-phosphate rearrangement, not any one organism's metabolic map.*

1. Executive Summary

The pentose phosphate pathway (PPP) is best understood not as a single linear route but as **two coupled biochemical modules** whose net direction is dictated by cellular boundary demand rather than by any fixed metabolic wiring diagram. The **oxidative branch** — glucose-6-phosphate dehydrogenase (G6PDH/Zwf), 6-phosphogluconolactonase (Pgl), and 6-phosphogluconate dehydrogenase (6PGDH/Gnd) — irreversibly converts glucose 6-phosphate (G6P) to ribulose 5-phosphate (Ru5P), paying out two molecules of NADPH and releasing CO₂. The **non-oxidative branch** — ribose-5-phosphate isomerase (Rpi), ribulose-phosphate 3-epimerase (Rpe), the thiamine-diphosphate (ThDP)-dependent transketolase, and the Schiff-base transaldolase — reversibly rearranges the five-carbon pool against fructose 6-phosphate (F6P) and glyceraldehyde 3-phosphate (G3P), supplying ribose 5-phosphate (R5P) for nucleotides and erythrose 4-phosphate (E4P) for aromatic amino acids.

The central mechanistic insight from this investigation is a **thermodynamic and directional asymmetry** between the two modules. Only the two dehydrogenase steps are truly obligatory and directional; the lactonase is a *protective accelerant* rather than a thermodynamically necessary step, and the entire non-oxidative branch operates near equilibrium and is freely reversible. As a direct consequence, the identical enzyme set can run in at least three distinct operating modes: (i) a forward, NADPH-generating oxidative mode; (ii) a reversed, ribose-synthesizing mode that draws R5P from glycolytic intermediates without spending the oxidative branch; and (iii) a cyclic mode in which oxidative and non-oxidative reactions merge

with glycolytic/gluconeogenic reactions to fine-tune redox output. The cyclic mode is exemplified by the **EDEMP cycle** of *Pseudomonas putida* KT2440 and by light/dark redox switching between the oxidative PPP and the reductive Calvin–Benson cycle in plant plastids.

The pathway's sugar-phosphate interconversion chemistry is **ancient and pre-enzymatic**: reconstituted Archean-ocean conditions catalyse many of the same reactions non-enzymatically, with Fe(II) acting as catalyst and cosubstrate and pH setting network topology. Against this deep conservation of chemistry sits a striking degree of *non-homologous convergence* in the protein families that later domesticated it — most clearly in ribose-5-phosphate isomerase, where two structurally unrelated families (RpiA and RpiB) independently catalyse the same reaction. The most important open questions concern quantitative flux control, the moonlighting/non-catalytic roles of PPP enzymes, and rigorous ancestral reconstruction of the expanded protein families.

2. Definition and Biological Boundaries

2.1 What is included

For the purposes of this review, the PPP comprises exactly the reactions that (a) oxidise G6P to Ru5P with NADPH production and (b) interconvert the resulting pentose phosphates with F6P and G3P. Concretely:

Oxidative branch (irreversible, directional):

Step	Enzyme (EC)	Reaction	Output
1	Glucose-6-phosphate dehydrogenase, Zwf (1.1.1.49/363)	G6P → 6-phospho-D-glucono-1,5-lactone	NAD(P)H
2	6-Phosphogluconolactonase, Pgl (3.1.1.31)	δ-lactone → 6-phospho-D-gluconate	(protective hydrolysis)
3	6-Phosphogluconate dehydrogenase, Gnd (1.1.1.44)	6-phosphogluconate → ribulose 5-phosphate + CO ₂	NADPH

Non-oxidative branch (near-equilibrium, reversible):

Step	Enzyme (EC)	Reaction
4	Ribose-5-phosphate isomerase, Rpi (5.3.1.6)	$\text{Ru5P} \rightleftharpoons \text{R5P}$
5	Ribulose-phosphate 3-epimerase, Rpe (5.1.3.1)	$\text{Ru5P} \rightleftharpoons \text{Xu5P}$
6	Transketolase, Tkt (2.2.1.1)	$\text{Xu5P} + \text{R5P} \rightleftharpoons \text{S7P} + \text{G3P}$
7	Transaldolase, Tal (2.2.1.2)	$\text{S7P} + \text{G3P} \rightleftharpoons \text{E4P} + \text{F6P}$
8	Transketolase, Tkt (2.2.1.1)	$\text{Xu5P} + \text{E4P} \rightleftharpoons \text{F6P} + \text{G3P}$

2.2 What should be treated separately

Several neighbouring processes are frequently conflated with the PPP but are mechanistically distinct and belong outside the boundary:

- **The Entner–Doudoroff (ED) pathway.** In many bacteria (notably *Pseudomonas*), G6P oxidation and 6-phosphogluconate share chemistry with the oxidative PPP but are then channelled through ED-specific enzymes (Edd/Eda) rather than 6PGDH. G6PDH is genuinely a *branch point* feeding either the reductive PPP or the ED route, so the two must be distinguished even though they share intermediates ([PMID: 33727391](#)).
- **The reductive Calvin–Benson cycle.** The non-oxidative PPP enzymes (transketolase, aldolase-type chemistry, and pentose interconversions) are *shared* with the reductive pentose-phosphate cycle in plastids, but the physiological direction and the presence/absence of active oxidative steps differ, and the two are reciprocally regulated ([PMID: 24079807](#)).
- **The phosphoketolase "bifid shunt."** Phosphoketolase is a ThDP-dependent relative of transketolase that cleaves Xu5P/F6P using inorganic phosphate to make acetyl-phosphate; it is a distinct enzyme with distinct chemistry ([PMID: 36974963](#)).
- **Downstream NADPH consumers** — the glutathione and thioredoxin reductase systems — are the *reason* the oxidative branch exists physiologically but are not part of the pathway itself.

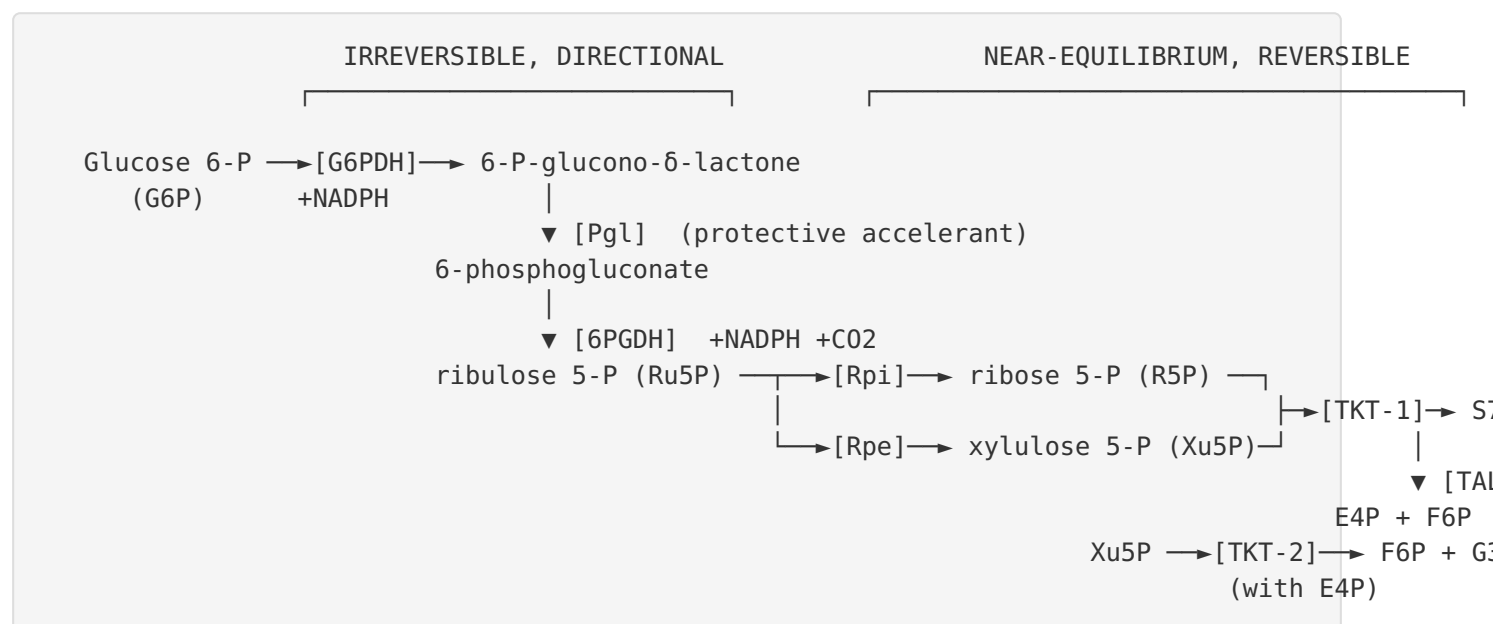
2.3 Competing definitions

The literature carries at least two implicit definitions. The classical textbook picture treats the PPP as a linear "shunt" off glycolysis with fixed stoichiometry. The systems/flux picture, supported here, treats it as a directionally plastic module whose operating mode is set by demand — the definition that best accommodates the EDEMP cycle and plastid redox gating.

3. Mechanistic Overview

3.1 The best current model: two modules, one directional gate

The core model that emerges from this investigation (**Finding F010**) is that the PPP is gated at exactly two points — the two dehydrogenases — and is otherwise a freely equilibrating carbon-shuffling network:



Two irreversible NADPH-producing steps commit carbon and set the forward direction; everything to the right of Ru5P is a reversible carbon economy that balances whatever the cell needs.

3.2 Obligatory vs. conditional vs. accessory steps

- **Obligatory and directional:** G6PDH and 6PGDH. These are the only two steps that are effectively irreversible under physiological conditions and thus define pathway directionality (**F001, F004, F010**).
- **Accessory (protective, not thermodynamically required):** 6-phosphogluconolactonase. NMR kinetics show that G6PDH produces exclusively the δ-(1→5) lactone, a highly electrophilic species that *can* hydrolyse spontaneously but can also isomerise to a "dead-end" γ-(1→4) form or react toxically with cellular nucleophiles. Pgl selectively accelerates

hydrolysis of the δ -form, channelling the reactive intermediate and preventing both the unproductive γ -lactone and cytotoxicity (**F008**; [PMID: 11457850](#)).

- **Conditional/reversible:** Rpi, Rpe, transketolase (both reactions), and transaldolase. These run in whichever direction the boundary metabolite concentrations dictate.

3.3 Catalytic chemistry of the non-oxidative enzymes

The non-oxidative branch is carried by two chemically distinct classes of carbon-transfer enzyme (**F002**):

- **Transketolase** is a **ThDP-dependent** transferase that moves two-carbon units. Catalysis proceeds through a reactive ThDP **ylide/carbene** intermediate, whose formation at physiological pH depends on dynamic protonation states within the active site ([PMID: 37748048](#)). The same enzyme performs both PPP two-carbon transfers (steps 6 and 8). Its stereoselective α -hydroxyketone chemistry has made it a workhorse of biocatalysis ([PMID: 41203356](#)), and bacterial transketolases can show unexpected cooperative kinetics and alternative substrate-binding sites ([PMID: 38145310](#)).
- **Transaldolase** transfers a three-carbon unit via a **lysine Schiff-base (class I aldolase)** mechanism and is entirely **independent of ThDP**.

This division of labour — ThDP for C2 transfer, a covalent lysine imine for C3 transfer — is a recurring theme and explains why the two enzymes have entirely different fold families despite acting consecutively.

4. Major Molecular Players and Active Assemblies

4.1 Glucose-6-phosphate dehydrogenase (G6PDH / Zwf)

G6PDH catalyses the **first committing step** and is a redox hub, not merely a gateway to Ru5P (**F001**). Two properties are central:

1. **Cofactor plasticity.** G6PDH isozymes differ in $\text{NADP}^+/\text{NAD}^+$ preference. In *Pseudomonas*, dual-cofactor isozymes exist, and cofactor specificity is a general principle underlying bacterial glycolytic strategy ([PMID: 33727391](#); [PMID: 36354357](#)). This lets the same reaction serve either biosynthetic NADPH supply or NADH-linked energy metabolism.
2. **Stability as the dominant failure mode.** In humans, G6PD deficiency is the most common enzymopathy (~400–500 million people). Detailed biochemistry of clinical variants shows

that most are **folding/stability defects, not catalytic defects (F005)**. For G6PD(Plymouth, G163D) and G6PD(Mahidol, G163S), $K_m(\text{NADP}^+)$, $K_m(\text{G6P})$ and k_{cat} match wild type, yet stability is markedly reduced — after 24 h at 37 °C, WT/Mahidol/Plymouth retained 58.3 / 27.0 / 3.9 % of activity, and NADP^+ binding stabilises the protein ([PMID: 17959407](#)). The bacterial enzyme is similarly stabilised by its substrate ([PMID: 31760039](#)), and small molecules that bridge the dimer interface can rescue oligomerisation and activity ([PMID: 31183991](#)).

4.2 6-Phosphogluconolactonase (Pgl)

A **protective accelerant** that channels the reactive δ -lactone (see §3.2, **F008**). Its dispensability in principle — but importance in practice — makes it the clearest "accessory but valuable" component of the pathway. The literature had long treated the lactonase as questionable precisely because the lactones were believed to hydrolyse spontaneously; the resolution is that the enzyme guards against accumulation of the toxic electrophilic δ -form ([PMID: 11457850](#)).

4.3 6-Phosphogluconate dehydrogenase (6PGDH / Gnd)

The **third PPP enzyme and a principal cellular NADPH generator** feeding thioredoxin/glutathione reductases (**F004**). It is an **obligate homodimer** with a Rossmann NADP^+ -binding domain. Structurally, its C-terminal tail penetrates the partner subunit: truncating the C-terminal 35–53 residues of yeast Gnd1 abolishes activity despite retained dimerisation ($K_m \approx 50 \mu\text{M}$ 6PG, $\approx 35 \mu\text{M}$ NADP^+) ([PMID: 17570834](#)). This intersubunit "arm" is an assembly constraint: the functional unit is the dimer, not the monomer.

4.4 Non-oxidative enzymes and non-homologous convergence

A striking feature is **lineage-specific family choice** for chemically identical reactions (**F009**):

- **Ribose-5-phosphate isomerase** exists as two structurally distinct, **non-homologous** classes, **RpiA** and **RpiB**. RpiB predominates in bacteria (with independent recruitments in fungi), yet both catalyse $\text{R5P} \rightleftharpoons \text{Ru5P}$ — a textbook case of convergent evolution to the same reaction ([PMID: 21995815](#)). RpiB's substrate specificity is itself evolutionarily labile and can be reshaped by gene loss ([PMID: 28362260](#)).
- **Ribulose-5-phosphate 3-epimerase** adopts a metal-dependent **TIM-barrel** fold, structurally related to epimerases such as UlaE from the anaerobic ascorbate pathway ([PMID: 18849419](#)).

The lesson for anyone asking "which family member best represents the ancestral role" is that there may be **no single ancestor** — different lineages solved the same isomerisation problem with unrelated scaffolds.

5. Evolutionary and Cell-Biological Variation

5.1 Deep, pre-enzymatic origins

The PPP's sugar-phosphate chemistry predates enzymes (**F003**). In reconstituted Archean-ocean mimetics, **29 reactions overlapping glycolysis and the PPP occur non-enzymatically**, including formation and interconversion of ribose 5-phosphate and erythrose 4-phosphate. **Fe(II)** acts as both catalyst and cosubstrate, binds sugar phosphates, and — together with pH — sets the network topology: alkaline conditions favour the PPP-like branch, while neutral/acidic conditions favour glycolysis-like reactivity ([PMID: 24771084](#); [PMID: 26824074](#)). This supports a **deep, plausibly pre-LUCA origin** of the pathway *topology*, with enzymes later domesticating a chemistry that already existed. Consistent with broader "gluconeogenic-first" arguments, the well-conserved trunk enzymes of central carbon metabolism are universally distributed, whereas upper-pathway steps derive from multiple independent gene families ([PMID: 15803666](#)).

5.2 Bacterial cyclic operation: the EDEMP cycle

Pseudomonas putida KT2440 provides the clearest example that the PPP is a *mode-switching* device (**F006**). ¹³C-flux analysis shows the strain **lacks a functional EMP glycolysis** and channels ~90 % of glucose (via gluconate/6-phosphogluconate) into the ED route, while ~10 % of triose phosphates are **recycled back to hexose phosphates**. This merges ED, gluconeogenic EMP, and PPP reactions into the **EDEMP cycle**, whose default state is a *slight catabolic overproduction of NADPH* — a redox buffer contributing to this organism's environmental robustness ([PMID: 26350459](#)). *P. putida* central metabolism further shows segregated flux architecture and GAPDH-homolog gatekeeping that tunes co-utilisation of substrates ([PMID: 30936206](#); [PMID: 41259139](#)), and the pathway is a practical engineering target for aromatic amino acid production from lignocellulosic sugars via the pentose phosphate pool ([PMID: 34403199](#)).

5.3 Plastid redox gating: avoiding a futile cycle

In photosynthetic plastids, the non-oxidative branch is **physically shared** with the reductive Calvin–Benson cycle, so running the oxidative branch in the light would create a futile cycle (F007). Chloroplast G6PDH — catalysing the first, rate-limiting oxidative step — is therefore **redox-regulated by thioredoxin**: a regulatory disulfide is reduced and the enzyme inactivated in the light, coordinating the oxidative PPP with reductive carbon fixation. Mechanistically, reduction of the regulatory disulfide changes substrate accessibility and NADPH-dependent cofactor binding via a conserved Arg residue (PMID: 24079807; PMID: 27717462). Plant plastidial G6PDH isoforms are also notoriously unstable and rapidly lose activity, complicating their study (PMID: 24161756).

5.4 Physiological states in animals

In mammals the oxidative branch is the redox linchpin of erythrocytes: G6PD deficiency renders red cells vulnerable to oxidative haemolysis after oxidant exposure (e.g., naphthalene, certain drugs) (PMID: 41763665; PMID: 39354868). Yet the phenotype is context-dependent and not uniformly deleterious — humanised G6PD-deficient mice can show *increased* exercise tolerance and improved cardiac mitochondrial function, cautioning against overgeneralising from the erythrocyte (PMID: 39514761). In cancer and immune biology, PPP-derived NADPH is a recurring node governing antioxidant capacity, redox-dependent cell death (ferroptosis, cuproptosis, disulfidptosis), and immune tolerance (PMID: 42190602; PMID: 42311140; PMID: 42397604; PMID: 42372728).

6. Constraints, Dependencies, and Failure Modes

6.1 Ordering constraints

The oxidative branch enforces a strict order because each step consumes the product of the previous one:



The two dehydrogenases are irreversible, so carbon cannot flow backward through the oxidative branch; ribose synthesised *without* NADPH production must therefore come from the **reversed non-oxidative branch** drawing on F6P and G3P. This is the key evidence that rules out an otherwise plausible "always-oxidative" route to ribose: cells that need R5P but not NADPH bypass the oxidative branch entirely.

6.2 Compartment- and state-specific exclusivity

- **Light vs. dark (plastids):** the oxidative branch and reductive carbon fixation are mutually exclusive in time, enforced by thioredoxin redox switching (§5.3).
- **Cofactor-specific routing:** whether G6PDH output supports biosynthesis or energy metabolism depends on the NADP⁺/NAD⁺ preference of the isozyme present (§4.1).
- **Substrate channelling:** the δ -lactone must be hydrolysed before it isomerises to the dead-end γ -form — a kinetic, not thermodynamic, constraint met by Pgl (§3.2).

6.3 Failure modes

Failure mode	Molecular basis	Consequence	Evidence
G6PDH instability	Folding/stability defects; loss of NADP ⁺ structural stabilisation	Oxidative haemolysis; most common human enzymopathy	PMID: 17959407, PMID: 31760039
Lactone toxicity	Loss of Pgl → δ -lactone reacts with nucleophiles or forms γ -dead-end	Toxic adducts, wasted carbon	PMID: 11457850
6PGDH assembly failure	C-terminal intersubunit arm truncation	Complete loss of activity despite intact dimer	PMID: 17570834
Redox miscoordination (plastid)	Failure to inactivate oxidative branch in light	Futile cycle with Calvin–Benson	PMID: 24079807

A recurring theme is that **stability and assembly — not catalytic competence — are the dominant failure points** for the oxidative-branch dehydrogenases.

7. Mechanistic Model / Interpretation (Synthesis)

Pulling the findings together, the PPP is most coherently described as a **directionally gated carbon-and-redox exchanger** with three operating modes:

Mode	Boundary demand	Oxidative branch	Non-oxidative branch	Net output
(i) Oxidative / NADPH	High NADPH need, low nucleotide need	ON (forward)	Forward: Ru5P → F6P + G3P	2 NADPH per G6P; CO ₂ released; carbon returned to glycolysis
(ii) Reversed / ribose-synthesis	High R5P need, low NADPH need	OFF	Reverse: F6P + G3P → pentose-P	R5P for nucleotides without spending oxidative branch
(iii) Cyclic (EDEMP / redox-buffer)	Redox robustness under stress	ON	Recycles trioses to hexoses	Net NADPH overproduction (<i>P. putida</i>)

The gate is set by the two irreversible dehydrogenases; the non-oxidative branch is a reversible "gearbox" that lets the cell take whatever carbon it needs from whichever direction is cheapest. Plastids add a fourth wrinkle by sharing the gearbox with reductive carbon fixation and switching the oxidative gate off in the light.

This model directly answers the review brief's central tension: the "known relationships among steps" (the linear feed-forward chain from G6PDH to the second transketolase reaction) are correct as a description of *substrate connectivity*, but they do not fix *direction*. Direction is an emergent property of boundary metabolite concentrations acting on a near-equilibrium network — which is why the same eight enzymes behave so differently in *Pseudomonas*, plastids, and mammalian erythrocytes.

8. Evidence Base

PMID	Contribution	Supports finding
33727391	G6PDH is first committing oxidative step; branch point to reductive PPP or ED; cofactor specificity principle	F001, F010
36354357	Oxidative step generates NAD(P)H; links to anabolism and oxidative-stress defense; dual-cofactor isozyme	F001
37748048	ThDP carbene intermediate in transketolase; dynamic protonation	F002
41203356	Transketolase ThDP mechanism and engineering	F002
24771084	Non-enzymatic PPP/glycolysis reactions in Archean-ocean mimetic; Fe(II) catalysis	F003
26824074	Iron- and pH-dependent conditional non-enzymatic network topology	F003
15803666	Gluconeogenic origin; conserved trunk vs. multi-family upper enzymes	F003
17570834	6PGDH is third PPP enzyme, main NADPH generator; obligate homodimer; C-terminal arm essential	F004
17959407	G6PD deficiency variants are folding/stability defects; NADP ⁺ stabilises	F005
31760039	Bacterial G6PDH stabilised by substrate	F005
31183991	Small-molecule G6PD activators bridge dimer interface	F005
26350459	EDEMP cycle in <i>P. putida</i> ; ~90% via ED, ~10% triose recycling; default NADPH overproduction	F006, F010
24079807	Chloroplast G6PDH thioredoxin redox regulation; light/dark coordination	F007, F010
27717462	Plastidic P2-G6PDH cysteine redox regulation and NADPH inhibition	F007
11457850	Pgl accelerates δ -lactone hydrolysis; prevents γ -dead-end and toxicity	F008, F010
21995815	Two non-homologous Rpi classes (RpiA/RpiB); convergent evolution	F009
18849419	Rpe/UlaE metal-dependent TIM-barrel fold	F009

PMID	Contribution	Supports finding
28362260	RpiB substrate specificity evolves via gene loss	F009
36974963	Phosphoketolase vs. transketolase distinction (boundary clarification)	Scope
38145310	Bacterial transketolase cooperativity and alternative sites	F002
39514761	G6PD deficiency can improve exercise/cardiac performance	Controversy

9. Controversies and Open Questions

- 1. Ancestral protein family for isomerisation.** Because RpiA and RpiB are non-homologous yet catalyse the same reaction, there is no single "best representative" of the ancestral isomerase. Whether one class is genuinely older, or whether the reaction was independently captured multiple times, remains unresolved ([PMID: 21995815](#); [PMID: 28362260](#)).
- 2. How linear is the "textbook" PPP?** Flux data from *P. putida* (EDEMP cycle) and plastids show that the linear map is often a poor description of real flux. The extent to which cyclic/reversed operation dominates in other organisms is largely unmeasured ([PMID: 26350459](#)).
- 3. Moonlighting and non-catalytic roles.** Glycolytic and PPP enzymes increasingly show non-enzymatic "moonlighting" functions in signalling and disease. Whether PPP enzymes beyond G6PD have physiologically important moonlighting roles is an open and largely correlative area ([PMID: 42311140](#)).
- 4. Organism-mixing in the redox-disease literature.** Much of the ferroptosis/cuproptosis/disulfidptosis literature links "PPP → NADPH → cell fate" using indirect, correlative evidence in cancer cell lines. These claims should not be read as mechanistic statements about the pathway's enzymology.
- 5. The physiological cost/benefit of the lactonase.** Pgl is dispensable in principle but protective in practice; quantifying how much toxic flux escapes in its absence, across organisms, is unresolved ([PMID: 11457850](#)).

6. **Why does G6PD deficiency sometimes help?** The counterintuitive exercise-tolerance and cardiac benefits in humanised deficient mice challenge the assumption that reduced oxidative-branch flux is uniformly harmful ([PMID: 39514761](#)).
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10. Limitations and Knowledge Gaps

- **Organism breadth.** Direct mechanistic evidence is concentrated in a few systems (yeast Gnd1, human G6PD variants, *Pseudomonas* flux, Arabidopsis/poplar plastid G6PDH, *E. coli*/fungal Rpi). Generalising quantitative flux behaviour to all lineages is not yet warranted.
 - **No primary data analysed here.** This review synthesises published biochemistry and structural work; it does not include new experimental measurements. The "findings" are literature-anchored inferences, not fresh assays.
 - **Flux control is under-quantified.** The three-mode model is qualitatively well supported but lacks organism-by-organism control coefficients that would say *how much* each gate contributes under defined conditions.
 - **Correlative disease literature.** The cancer/immunometabolism papers link PPP-derived NADPH to cell-fate decisions largely by correlation; these should be treated as motivating context, not mechanism.
 - **Ancestral reconstruction is incomplete.** The non-homology of RpiA/RpiB means the "ancestral isomerase" question cannot currently be answered from sequence data alone.
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11. Proposed Follow-up Experiments / Actions

1. **Quantitative multi-organism flux atlas.** Apply ^{13}C -metabolic flux analysis across phylogenetically diverse microbes (beyond *P. putida*) under matched conditions to measure how often the reversed and cyclic modes dominate, and to estimate control coefficients for the two dehydrogenase gates.
2. **Pgl channelling assay in vivo.** Use isotope-labelled δ -lactone and mass spectrometry in Pgl-knockout vs. wild-type cells to quantify how much reactive lactone escapes to nucleophile adducts or the γ -dead-end, directly testing the "protective accelerant" model.

3. **Ancestral sequence reconstruction of transketolase and 6PGDH.** Resurrect and characterise inferred ancestral enzymes to test whether ancestral forms were cofactor-generalist and whether the obligate 6PGDH dimer arm is ancestral or derived.
4. **Systematic G6PD variant stability screen.** Extend the folding-defect model with high-throughput thermostability and NADP⁺-rescue assays across clinical variants to build a predictive stability–phenotype map, and test small-molecule dimer-interface activators (PMID: 31183991) as chemical chaperones.
5. **Plastid futile-cycle test.** Engineer thioredoxin-insensitive G6PDH in plants and measure whether light-driven futile cycling measurably penalises carbon fixation, quantifying the physiological value of redox gating.
6. **RpiA vs. RpiB fitness swap.** Complement an RpiB-dependent bacterium with RpiA (and vice versa) to test whether family choice is neutral or lineage-adaptive.

12. Key References

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- *Evolution of substrate specificity in a retained enzyme driven by gene loss (RpiA/RpiB).* PMID: 28362260
- *Dynamic Protonation States Underlie Carbene Formation in ThDP-Dependent Enzymes.* PMID: 37748048
- *Distribution and phylogenies of EMP enzymes... support a gluconeogenic origin of metabolism.* PMID: 15803666

13. Conclusion

The pentose phosphate pathway is neither a simple linear shunt nor a single cycle: it is a **two-module, directionally gated central-carbon device**. Two irreversible dehydrogenase steps (G6PDH, 6PGDH) commit carbon and produce NADPH; a protective lactonase channels a reactive intermediate; and a reversible, near-equilibrium non-oxidative gearbox (Rpi, Rpe, transketolase, transaldolase) rebalances pentose phosphates against F6P and G3P. Because only the gate is directional, the same enzymes run forward for NADPH, backward for ribose, and cyclically for redox buffering — as demonstrated by the EDEMP cycle in *Pseudomonas putida* and by light/dark redox switching in plastids. The pathway's chemistry is ancient enough to have non-enzymatic, iron-catalysed prebiotic counterparts, yet its enzymes show repeated non-homologous convergence, most vividly in the two unrelated ribose-5-phosphate isomerase families. The largest open questions are quantitative flux control, enzyme moonlighting, and rigorous ancestral reconstruction of the expanded families.